Lesson 8.3 — Sampling and Surveys

Big idea: How you collect data is at least as important as how you analyze it. A bad sample produces a confident wrong answer; a good sample produces an honest estimate of the population.

Quick review

Random sample: every member of the population has a known, nonzero chance of being chosen. Removes bias.

Common bias sources: voluntary response, convenience sampling, undercoverage (some groups underrepresented), nonresponse, leading questions.

Sampling methods: simple random, stratified (split into groups, sample within), cluster (sample whole groups), systematic (every kth).

Bigger sample ≠ better if biased. Increasing a biased sample makes it more confidently wrong.

Worked example

Worked example. A radio show invites listeners to call in with their opinion on a tax. 80% support it. Is this an unbiased estimate?

No — voluntary response bias. People with strong opinions (especially against) tend to call in. The 80% statistic estimates only those who chose to call.

Warm-ups

1. Name the bias: A reporter only interviews people coming out of a gym, asking how often they exercise.
2. Name the bias: A magazine surveys its own subscribers about politics, then claims to know the country's opinion.
3. Is a sample of 5,000 always better than a sample of 500?

Practice

1. Identify the sampling method: a school surveys 5 students chosen at random from each grade level.
2. Identify the sampling method: surveying every 10th student leaving a cafeteria.
3. A poll question reads: "Don't you agree that the unfair tax should be repealed?" Identify the bias.

Challenge

1. A school wants to estimate the proportion of students who walk to school. They survey students entering the parking lot. What bias appears, and how would you redesign the sample?
2. If a stratified sample is taken (with proportionate sampling within strata), explain why it can be more precise than a simple random sample of the same size.

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Answer key — Lesson 8.3

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. Convenience / selection bias (gym-goers exercise more than the population).
2. Coverage bias / undercoverage (subscribers are not a random slice of the country).
3. Not necessarily — bias matters more than size. A bigger biased sample is just confidently wrong.

Practice

1. Stratified random sampling.
2. Systematic sampling.
3. Leading question (wording prejudges the answer).

Challenge

1. Coverage bias: students who walk would never be in the parking lot. Better: stratify or randomly sample students at school entrances or from the official roster.

2. If groups (strata) differ on the variable being measured, sampling proportionally from each captures the within-group variation while reducing between-group sampling noise. The result is a smaller standard error than a simple random sample of the same total size.

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